

Análisis de resultados — B1-B4, it1-it130

Generado: 2026-04-13T02:36:30.233519

Combinaciones totales: 315

Combinaciones significativas: 129

Resumen global

Resumen global de métodos

Generado: 2026-04-10T18:26:44.791941Z

Top métodos (por media MAE)

method	count	mean	std	median
---	---	---	---	---
spline_temporal	32	1.1654829380302845e-05	2.7209503884245993e-06	1.1134885641167803e-05
gsmooth	5274	0.09259058115502374	0.09648843752920808	0.0652929243435412
idw	5406	0.09545196817289028	0.0991223097162261	0.0660994520829936
gsp	3173	0.11583293477015401	0.10817258236108003	0.088259445073733
directed_tv	2560	0.12060995383095234	0.17043577359727477	0.1181534984305608
linear	3512	0.120845823006665	0.11800635226800493	0.0942564113430665
wavelet_temporal	2550	0.12231655323920902	0.10331373770962676	0.123798954736391
spherical_spline	3306	0.13465792977927354	0.14202871587133703	0.1021720677531056
rbfi_tps	3306	0.14401542249789828	0.14919248365159427	0.0950698397827097
rbfi_mq	3306	0.1607660111084691	0.1705946737790419	0.1076814030559054

Top-3 apariciones por bloque

method	top3_block_count
---	---
tv	2420
trss	2337
temporal_laplacian	2297
graph_time_tikhonov	2278
mean	2253
tikhonov	2235
nearest	2073
idw	756
gsmooth	752
linear	752
directed_tv	752
heat_diffusion_temporal	744

Resumen global

```
|wavelet_temporal|743|
|spherical_spline|610|
|puy|590|
|spline_surface|590|
|rbfi_tps|590|
|rbfi_mq|590|
|bgsrp|589|
|sobolev|572|
|gsp|534|
|spline_temporal|32|
```

Top-3 apariciones por dataset

```
|method|top3_dataset_count|
|---|---|
|tv|14|
|mean|11|
|trss|10|
|gsmooth|7|
|graph_time_tikhonov|3|
|nearest|3|
|directed_tv|3|
|spherical_spline|2|
|idw|2|
|linear|1|
|wavelet_temporal|1|
|heat_diffusion_temporal|1|
|tikhonov|1|
|temporal_laplacian|1|
```

Estadísticas por combinación (graph + dataset + missing_ratio)

Se ha calculado la media de MAE por método para cada combinación (graph, dataset, missing_ratio), seleccionando el *mejor* (menor MAE) y el segundo mejor para comparar. Para cada combinación se incluye:

- `best\_method`, `best\_mean\_mae`, `best\_n`, `ci\_lower`, `ci\_upper`

Resumen global

- `second\_method`, `second\_mean\_mae`, `second\_n`
- `delta\_mean` (segundo - mejor)
- `p\_raw`, `p\_adj` (p-valor de Mann-Whitney y p ajustado por Holm), `significant`
- `rank\_biserial` (efecto, rango-biserial)

Resumen rápido:

- Combinaciones evaluadas: 315
- Combinaciones con mejora significativa ($p_{adj} < 0.05$): 129
- Top métodos reportados como "mejor": `trss` (107), `tv` (54), `heat\_diffusion\_temporal` (34), `directed\_tv` (31), `mean` (28)

Archivos con resultados completos:

- Tabla completa (CSV): Thesis-Copilot-Toolkit/results/analysis/batches/best_method_by_combo_stats.csv
- Tabla resumen (mejor por combinación): Thesis-Copilot-Toolkit/results/analysis/batches/best_method_by_combo.csv
- Libro Excel con todas las hojas (descargable): Thesis-Copilot-Toolkit/results/analysis/batches/summary_report.xlsx
- Dashboard interactivo (index): Thesis-Copilot-Toolkit/results/analysis/batches/interactive/index.html

Notas:

- Los IC95 se calcularon por bootstrap sobre la media del MAE (2000 repeticiones cuando hay suficiente muestra).
- La prueba usada para comparar mejor vs segundo fue Mann-Whitney U (no paramétrica). Los p-valores se ajustaron con Holm.
- Para combinaciones con muestras pequeñas ($n < 3$) no se reportaron p-valores.

Si quieres, puedo:

- Exportar únicamente las 129 combinaciones significativas a un XLSX separado.
- Añadir una figura (heatmap) que muestre solo combinaciones significativas por `best\_method`.
- Subir `summary\_report.xlsx` a un servicio (Google Drive, transfer.sh, etc.) y devolver un enlace

Resumen global

descargable.

Resumen legible: mejores métodos

Input file: best_method_by_combo_stats.csv

Total combinations: 315

Significant combinations: 129

Top methods by count:

- trss: 107
- tv: 54
- heat_diffusion_temporal: 34
- directed_tv: 31
- mean: 28
- spherical_spline: 24
- linear: 8
- nearest: 8
- gsmooth: 6
- gsp: 4
- idw: 3
- rbfi_tps: 3
- temporal_laplacian: 2
- wavelet_temporal: 2
- graph_time_tikhonov: 1

Top 10 combos by delta_mean (second_mean - best_mean):

- gaussian_sigma1.0 | iv100hz_mat | 0.4 -> tv (delta=12.401090303957488)
- knn_k3 | iv100hz_mat | 0.2 -> trss (delta=9.549269867852187)
- knn_k3 | iv100hz_mat | 0.3 -> trss (delta=7.470274733956003)
- gaussian_sigma1.0 | iv100hz_mat | 0.3 -> tv (delta=7.175423791513204)
- knn_k3 | iv100hz_mat | 0.4 -> trss (delta=5.966905123785836)
- gaussian_sigma1.0 | iv100hz_mat | 0.2 -> tv (delta=3.4666914353750364)
- gaussian_sigma1.0 | iv100hz_mat | 0.1 -> tv (delta=1.8049861163888803)
- knn_k3 | iv100hz_mat | 0.1 -> trss (delta=1.360573863855027)
- knn_k5 | iv100hz_mat | 0.3 -> trss (delta=0.7920825289618065)
- knn_k5 | iv100hz_mat | 0.2 -> tv (delta=0.581036898496798)

Top method counts

best_method	count
trss	107
tv	54
heat_diffusion_temporal	34
directed_tv	31
mean	28
spherical_spline	24
linear	8
nearest	8
gsmooth	6
gsp	4
idw	3
rbfi_tps	3
temporal_laplacian	2
wavelet_temporal	2
graph_time_tikhonov	1

Top20 combos por delta

graph	dataset	single	best_test	mean	best_n	best_std	lower	upper	second	mean	second	delta	mean	p_raw	p_adj	significant	k_biseria
gauss	iv100	0.4	tv	61.66	2	1.188	60.82	62.50	trss	74.06	2	1.003	12.40	nan	nan	False	nan
knn	iv100	0.2	trss	36.77	118	15.14	33.97	39.55	tv	46.32	118	12.07	9.549	2.948	8.520	True	0.6367
knn	iv100	0.3	trss	65.71	25	0.917	65.36	66.07	mean	73.18	25	0.708	7.470	1.304	3.601	True	1.0
gauss	iv100	0.3	tv	48.52	25	2.169	47.72	49.29	trss	55.70	25	0.735	7.175	1.304	3.614	True	1.0
knn	iv100	0.4	trss	88.19	2	0.291	87.98	88.39	mean	94.15	2	1.017	5.966	nan	nan	False	nan
gauss	iv100	0.2	tv	26.35	27	1.979	25.66	27.14	trss	29.82	27	0.584	3.466	1.436	3.145	True	0.8984
gauss	iv100	0.1	tv	12.42	20	0.735	12.13	12.73	trss	14.23	20	0.449	1.804	8.725	1.666	True	0.99
knn	iv100	0.1	trss	16.15	20	0.385	15.99	16.32	tv	17.51	20	1.043	1.360	8.300	0.001	True	0.8254
knn	iv100	0.3	trss	62.75	5	1.007	61.98	63.52	tv	63.54	5	5.371	0.792	0.841	1.0	False	0.12
knn	iv100	0.2	tv	32.65	5	3.385	30.40	35.47	trss	33.24	5	0.619	0.581	0.309	1.0	False	0.4399
gauss	iris_g	0.4	trss	0.228	2	0.007	0.223	0.233	tv	0.266	2	0.008	0.038	nan	nan	False	nan
knn	iris_g	0.2	trss	0.107	114	0.029	0.102	0.113	tv	0.134	114	0.023	0.027	6.337	1.837	True	0.7199
knn	iris_g	0.3	trss	0.103	5	0.009	0.096	0.111	tv	0.127	5	0.011	0.023	0.015	1.0	False	0.92
knn	iris_g	0.2	trss	0.103	5	0.009	0.096	0.111	tv	0.127	5	0.011	0.023	0.015	1.0	False	0.92
gauss	iris_g	0.2	trss	0.102	17	0.008	0.098	0.106	tv	0.123	17	0.011	0.020	2.454	0.004	True	0.8477
knn	iris_g	0.3	trss	0.105	21	0.009	0.101	0.109	tv	0.126	21	0.011	0.020	8.844	0.001	True	0.8004
gauss	iris_g	0.3	trss	0.102	15	0.008	0.097	0.106	tv	0.122	15	0.011	0.020	0.000	0.025	True	0.8139
knn	iris_g	0.1	trss	0.105	28	0.008	0.102	0.108	tv	0.126	28	0.011	0.020	1.200	2.220	True	0.8239
gauss	iris_g	0.1	trss	0.102	10	0.008	0.097	0.107	tv	0.122	10	0.012	0.020	0.002	0.387	False	0.8
kalofc	synth	0.1	trss	0.476	9	0.022	0.462	0.489	graph	0.491	9	0.028	0.014	0.185	1.0	False	0.3827

Mejores combinaciones (ordenadas) — primeras filas

graph	data	single	best_test	mean	best_n	best_std	lower	upper	cmd	mean	second	second	delta	p_raw	p_adj	significant	k	bise
gauss	bci_cc	0.1	direct	3.563	10	4.406	3.317	3.839	tv	3.645	10	2.575	8.246	0.427	1.0	False	0.219	
gauss	bci_cc	0.3	direct	4.889	10	5.234	4.572	5.189	tv	4.906	10	3.866	1.664	0.969	1.0	False	-0.02	
knn	bci_cc	0.1	direct	3.635	13	5.163	3.383	3.933	trss	3.636	13	4.772	4.148	0.959	1.0	False	0.017	
knn	mne	0.1	direct	1.226	4	6.302	1.176	1.277	heat	1.335	4	1.542	1.085	0.485	1.0	False	0.375	
knn	mne	0.2	direct	1.479	4	1.530	1.351	1.596	heat	1.527	4	2.210	4.741	1.0	1.0	False	0.0	
knn	mne	0.4	direct	2.007	5	3.670	1.743	2.328	heat	2.132	5	2.651	1.248	0.420	1.0	False	0.36	
gauss	synth	0.3	direct	0.199	18	2.856	0.199	0.199	heat	0.207	18	0.0	0.008	3.683	9.613	True	1.0	
kalofo	synth	0.3	direct	0.200	18	0.0	0.200	0.200	mean	0.212	18	0.0	0.012	3.683	9.503	True	1.0	
knn	synth	0.3	direct	0.202	19	2.851	0.202	0.202	heat	0.209	19	0.0	0.006	1.312	3.529	True	1.0	
knng	synth	0.3	direct	0.200	17	0.0	0.200	0.200	heat	0.207	17	0.0	0.006	1.036	2.362	True	1.0	
vknn	synth	0.3	direct	0.200	17	0.0	0.200	0.200	wave	0.208	17	0.0	0.008	1.036	2.300	True	1.0	
aew	synth	0.4	direct	0.255	15	0.0	0.255	0.255	trss	0.262	15	5.745	0.007	8.265	1.603	True	1.0	
gauss	synth	0.3	direct	0.202	17	0.0	0.202	0.202	heat	0.203	17	0.0	0.001	1.036	2.414	True	1.0	
gauss	synth	0.4	direct	0.269	17	0.0	0.269	0.269	heat	0.276	17	0.0	0.007	1.036	2.404	True	1.0	
kalofo	synth	0.3	direct	0.202	18	0.0	0.202	0.202	mean	0.203	18	0.0	0.001	3.683	9.392	True	1.0	
kalofo	synth	0.4	direct	0.271	18	0.0	0.271	0.271	idw	0.281	18	0.0	0.009	3.683	9.355	True	1.0	
knn	synth	0.4	direct	0.269	18	5.712	0.269	0.269	heat	0.272	18	5.712	0.003	3.683	9.171	True	1.0	
vknn	synth	0.4	direct	0.274	16	0.0	0.274	0.274	heat	0.275	16	0.0	0.000	2.922	5.903	True	1.0	
gauss	synth	0.1	direct	0.078	19	0.0	0.078	0.078	heat	0.081	19	0.0	0.002	1.312	3.608	True	1.0	
gauss	synth	0.2	direct	0.118	19	0.0	0.118	0.118	heat	0.120	19	1.425	0.002	1.312	3.595	True	1.0	
gauss	synth	0.3	direct	0.197	20	2.847	0.197	0.197	heat	0.198	20	2.847	0.000	4.682	1.329	True	1.0	
kalofo	synth	0.1	direct	0.081	19	0.0	0.081	0.081	idw	0.083	19	1.425	0.001	1.312	3.581	True	1.0	
kalofo	synth	0.2	direct	0.120	19	2.851	0.120	0.120	gsmo	0.124	19	2.851	0.003	1.312	3.568	True	1.0	
kalofo	synth	0.3	direct	0.197	20	0.0	0.197	0.197	mean	0.201	20	2.847	0.004	4.682	1.320	True	1.0	
kalofo	synth	0.4	direct	0.236	20	2.847	0.236	0.236	mean	0.238	20	2.847	0.002	4.682	1.315	True	1.0	
knn	synth	0.1	direct	0.068	21	0.021	0.058	0.075	heat	0.072	21	0.022	0.003	4.588	8.304	True	0.823	
knn	synth	0.1	direct	0.075	18	0.0	0.075	0.075	heat	0.080	18	0.0	0.004	3.683	9.134	True	1.0	
knn	synth	0.2	direct	0.119	18	2.856	0.119	0.119	heat	0.120	18	2.856	0.000	3.683	9.097	True	1.0	
knn	synth	0.3	direct	0.198	19	0.0	0.198	0.198	heat	0.198	19	0.0	0.000	1.312	3.516	True	1.0	
knng	synth	0.1	direct	0.076	17	0.0	0.076	0.076	heat	0.080	17	0.0	0.004	1.036	2.352	True	1.0	

Mejores combinaciones (ordenadas) — primeras filas

graph	dataset	single	best_test	mean	best_n	best_std	lower	upper	second	mean	second	third	delta	mean	p_raw	p_adj	significant	k_bise
vknn	synth	0.1	direct	0.076	17	0.0	0.076	0.076	heat	0.080	17	0.0	0.004	1.036	2.290	True	1.0	
kalofc	synth	0.4	graph	0.528	3	0.012	0.513	0.536	tv	0.539	3	0.009	0.010	0.4	1.0	False	0.555	
aew	mne	0.1	gsmo	2.807	60	1.030	2.543	3.064	gsp	2.824	20	1.040	1.688	0.880	1.0	False	0.023	
knn	mne	0.1	gsmo	2.833	30	1.064	2.464	3.212	tv	2.835	30	1.023	1.468	0.784	1.0	False	0.042	
knng	mne	0.1	gsmo	2.833	60	1.055	2.561	3.094	tv	2.835	60	1.015	1.431	0.691	1.0	False	0.042	
knng	synth	0.2	gsmo	0.132	16	0.0	0.132	0.132	heat	0.132	16	0.0	2.823	2.922	6.108	True	1.0	
nnk__	synth	0.1	gsmo	0.065	18	0.0	0.065	0.065	heat	0.065	18	0.0	0.000	3.683	8.876	True	1.0	
vknn	synth	0.2	gsmo	0.132	16	0.0	0.132	0.132	heat	0.132	16	0.0	5.368	2.922	5.961	True	1.0	
kalofc	physio	0.4	gsp	1.191	30	1.391	1.142	1.241	gsmo	1.252	31	1.552	6.108	0.023	1.0	False	0.337	
knn__	physio	0.1	gsp	1.665	77	1.437	1.634	1.699	gsmo	1.831	78	1.678	1.653	2.194	4.762	True	0.520	
knng	physio	0.1	gsp	1.692	5	2.367	1.692	1.692	trss	1.700	6	0.0	8.046	0.002	0.319	False	1.0	
vknn	physio	0.1	gsp	1.692	5	2.367	1.692	1.692	trss	1.700	6	2.319	7.995	0.002	0.317	False	1.0	
kalofc	bci_cc	0.1	heat	3.069	1	nan	nan	nan	trss	3.254	1	nan	1.848	nan	nan	False	nan	
kalofc	bci_cc	0.3	heat	4.125	1	nan	nan	nan	trss	4.179	1	nan	5.464	nan	nan	False	nan	
kalofc	bci_cc	0.4	heat	5.213	1	nan	nan	nan	direct	5.344	1	nan	1.307	nan	nan	False	nan	
kalofc	mne	0.1	heat	1.349	1	nan	nan	nan	direct	1.584	1	nan	2.356	nan	nan	False	nan	
kalofc	mne	0.2	heat	1.625	1	nan	nan	nan	wave	1.766	1	nan	1.408	nan	nan	False	nan	
kalofc	mne	0.3	heat	1.850	1	nan	nan	nan	wave	1.869	1	nan	1.860	nan	nan	False	nan	
knn__	mne	0.3	heat	1.776	6	2.268	1.614	1.944	direct	1.865	6	2.933	8.906	0.818	1.0	False	0.111	
knn__	synth	0.3	heat	0.194	20	0.044	0.174	0.204	direct	0.197	20	0.045	0.003	4.342	8.641	True	0.9	
gauss	synth	0.2	heat	0.134	17	0.0	0.134	0.134	tv	0.134	17	0.0	6.842	1.036	2.424	True	1.0	
knn__	synth	0.1	heat	0.061	19	0.013	0.055	0.064	gsmo	0.061	19	0.013	0.000	9.924	1.875	True	0.900	
knn__	synth	0.3	heat	0.193	19	0.045	0.172	0.203	mean	0.193	19	0.044	0.000	9.924	1.865	True	0.900	
knn__	synth	0.4	heat	0.259	19	0.061	0.231	0.273	direct	0.265	19	0.062	0.005	1.188	2.211	True	0.894	
knng	synth	0.1	heat	0.065	16	0.0	0.065	0.065	gsmo	0.065	16	0.0	0.000	2.922	6.137	True	1.0	
knng	synth	0.3	heat	0.201	16	0.0	0.201	0.201	gsmo	0.202	16	0.0	0.000	2.922	6.078	True	1.0	
knng	synth	0.4	heat	0.273	16	0.0	0.273	0.273	direct	0.274	16	0.0	0.001	2.922	6.049	True	1.0	
nnk__	synth	0.3	heat	0.202	18	0.0	0.202	0.202	mean	0.203	18	0.0	0.001	3.683	8.803	True	1.0	
nnk__	synth	0.4	heat	0.273	18	5.712	0.273	0.273	idw	0.281	18	0.0	0.007	3.683	8.766	True	1.0	
aew	synth	0.2	heat	0.119	15	4.309	0.119	0.119	trss	0.120	15	5.745	0.000	8.265	1.587	True	1.0	

Mejores combinaciones (ordenadas) — primeras filas

graph	dataset	single	best_test	mean	best_n	best_std	lower	upper	second	mean	second	cond	delta	mean	p_raw	p_adj	significant	k_bise
aew__	synth	0.3	heat__	0.197	16	0.0	0.197	0.197	trss	0.199	16	0.0	0.002	2.922	6.224	True	1.0	
gauss__	synth	0.4	heat__	0.235	20	0.0	0.235	0.235	trss	0.236	20	5.695	0.001	4.682	1.325	True	1.0	
knn__	synth	0.2	heat__	0.108	21	0.034	0.092	0.119	direct	0.110	21	0.034	0.001	4.588	8.258	True	0.823	
knn__	synth	0.3	heat__	0.181	22	0.056	0.155	0.199	spher	0.183	21	0.058	0.001	2.871	5.255	True	0.826	
knn__	synth	0.4	heat__	0.215	22	0.067	0.183	0.236	spher	0.216	21	0.068	0.001	2.871	5.226	True	0.826	
knn__	synth	0.4	heat__	0.235	19	0.0	0.235	0.235	tv	0.236	19	5.703	0.001	1.312	3.503	True	1.0	
knng__	synth	0.2	heat__	0.119	17	0.0	0.119	0.119	direct	0.121	17	0.0	0.001	1.036	2.341	True	1.0	
knng__	synth	0.3	heat__	0.198	18	0.0	0.198	0.198	direct	0.201	18	0.0	0.002	3.683	9.061	True	1.0	
knng__	synth	0.4	heat__	0.235	18	0.0	0.235	0.235	tv	0.236	18	0.0	0.000	3.683	9.024	True	1.0	
nnk__	synth	0.1	heat__	0.080	19	1.425	0.080	0.080	trss	0.081	19	0.0	0.001	1.312	3.490	True	1.0	
nnk__	synth	0.2	heat__	0.119	19	1.425	0.119	0.119	trss	0.124	19	0.0	0.004	1.312	3.476	True	1.0	
nnk__	synth	0.3	heat__	0.197	20	0.0	0.197	0.197	mean	0.201	20	2.847	0.004	4.682	1.311	True	1.0	
nnk__	synth	0.4	heat__	0.236	20	0.0	0.236	0.236	mean	0.238	20	2.847	0.001	4.682	1.306	True	1.0	
vkng__	synth	0.2	heat__	0.119	17	1.430	0.119	0.119	direct	0.120	17	1.430	0.000	1.036	2.279	True	1.0	
vkng__	synth	0.3	heat__	0.199	18	0.0	0.199	0.199	direct	0.200	18	0.0	0.000	3.683	8.729	True	1.0	
vkng__	synth	0.4	heat__	0.235	18	0.0	0.235	0.235	tv	0.236	18	5.712	0.000	3.683	8.692	True	1.0	
kalofc__	physic	0.3	idw	8.746	26	5.727	8.532	8.953	rbfi_tf	8.795	25	7.303	4.874	0.359	1.0	False	0.150	
knn__	physic	0.3	idw	8.708	87	7.212	8.563	8.865	rbfi_tf	9.010	77	7.810	3.019	0.027	1.0	False	0.199	
knn__	synth	0.3	idw	0.173	54	0.034	0.163	0.181	trss	0.177	72	0.032	0.004	0.229	1.0	False	0.125	
aew__	physic	0.2	linear	3.315	6	0.0	3.315	3.315	trss	3.976	6	0.0	6.606	0.001	0.194	False	1.0	
gauss__	physic	0.2	linear	3.292	25	2.482	3.199	3.385	gsmo	4.419	25	3.661	1.126	1.278	3.555	True	1.0	
kalofc__	physic	0.2	linear	3.335	25	2.529	3.236	3.431	gsp	4.529	24	2.919	1.193	1.925	5.083	True	1.0	
knn__	physic	0.2	linear	3.364	87	3.018	3.301	3.430	idw	4.923	87	4.009	1.558	4.675	1.369	True	1.0	
knn__	physic	0.2	linear	3.315	6	0.0	3.315	3.315	gsmo	4.378	6	0.0	1.062	0.001	0.193	False	1.0	
knng__	physic	0.2	linear	3.315	6	0.0	3.315	3.315	gsmo	4.435	6	9.278	1.120	0.001	0.190	False	1.0	
nnk__	physic	0.2	linear	3.315	7	0.0	3.315	3.315	tv	4.438	7	0.0	1.122	0.000	0.066	False	1.0	
vkng__	physic	0.2	linear	3.315	6	0.0	3.315	3.315	gsmo	4.470	6	0.0	1.155	0.001	0.188	False	1.0	
gauss__	movie	0.1	mean	0.016	10	0.001	0.015	0.017	tv	0.016	10	0.001	9.541	0.569	1.0	False	0.16	
knn__	movie	0.1	mean	0.016	26	0.002	0.016	0.017	tv	0.016	26	0.002	4.671	0.748	1.0	False	0.053	
knn__	movie	0.2	mean	0.035	44	0.002	0.034	0.036	tv	0.036	90	0.003	0.000	0.810	1.0	False	0.025	

Mejores combinaciones (ordenadas) — primeras filas

graph	dataset	single	best_test	mean	best_n	best_std	lower	upper	second	mean	second	cond	delta	mean	p_raw	p_adj	significant	k_bise
aew_	synth	0.1	mean	0.058	16	0.0	0.058	0.058	idw	0.062	16	0.0	0.003	2.922	6.312	True	1.0	
aew_	synth	0.2	mean	0.120	16	0.0	0.120	0.120	gsmo	0.128	16	0.0	0.007	2.922	6.283	True	1.0	
aew_	synth	0.4	mean	0.261	15	0.0	0.261	0.261	wave	0.269	15	0.0	0.008	8.265	1.628	True	1.0	
gauss	synth	0.1	mean	0.058	18	7.140	0.058	0.058	tv	0.059	18	7.140	0.000	3.683	9.687	True	1.0	
gauss	synth	0.4	mean	0.261	17	0.0	0.261	0.261	gsmo	0.262	17	0.0	0.001	1.036	2.435	True	1.0	
kalofc	synth	0.1	mean	0.058	18	7.140	0.058	0.058	tv	0.058	18	7.140	3.849	3.683	9.576	True	1.0	
kalofc	synth	0.2	mean	0.120	18	1.428	0.120	0.120	tv	0.120	18	1.428	1.441	3.683	9.539	True	1.0	
kalofc	synth	0.4	mean	0.261	17	0.0	0.261	0.261	tv	0.261	17	0.0	7.384	1.036	2.393	True	1.0	
knn_	synth	0.1	mean	0.056	20	0.012	0.050	0.058	idw	0.059	20	0.012	0.003	3.653	7.343	True	0.905	
knn_	synth	0.2	mean	0.115	20	0.025	0.103	0.120	gsmo	0.122	20	0.027	0.007	4.342	8.684	True	0.9	
knn_	synth	0.1	mean	0.058	19	0.0	0.058	0.058	tv	0.061	19	1.425	0.002	1.312	3.555	True	1.0	
knn_	synth	0.2	mean	0.120	19	0.0	0.120	0.120	tv	0.121	19	0.0	0.000	1.312	3.542	True	1.0	
knn_	synth	0.4	mean	0.261	18	0.0	0.261	0.261	tv	0.261	18	5.712	0.000	3.683	9.318	True	1.0	
knng_	synth	0.1	mean	0.058	17	7.152	0.058	0.058	idw	0.062	17	0.0	0.003	1.036	2.383	True	1.0	
knng_	synth	0.2	mean	0.120	17	1.430	0.120	0.120	gsmo	0.128	17	0.0	0.007	1.036	2.373	True	1.0	
nnk_	synth	0.1	mean	0.058	18	7.140	0.058	0.058	idw	0.062	18	0.0	0.003	3.683	8.987	True	1.0	
nnk_	synth	0.2	mean	0.120	18	1.428	0.120	0.120	gsmo	0.127	18	0.0	0.007	3.683	8.950	True	1.0	
vkng_	synth	0.1	mean	0.058	17	7.152	0.058	0.058	idw	0.062	17	0.0	0.003	1.036	2.321	True	1.0	
vkng_	synth	0.2	mean	0.120	17	1.430	0.120	0.120	tv	0.127	17	0.0	0.006	1.036	2.310	True	1.0	
vkng_	synth	0.4	mean	0.261	16	0.0	0.261	0.261	tv	0.262	16	0.0	0.000	2.922	6.020	True	1.0	
kalofc	synth	0.2	mean	0.134	18	0.0	0.134	0.134	tv	0.134	18	0.0	6.465	3.683	9.429	True	1.0	
knn_	synth	0.2	mean	0.127	19	0.028	0.114	0.134	idw	0.127	19	0.029	0.000	1.188	2.223	True	0.894	
knn_	synth	0.1	mean	0.065	18	0.0	0.065	0.065	heat_	0.065	18	1.428	2.513	3.683	9.282	True	1.0	
knn_	synth	0.2	mean	0.134	18	0.0	0.134	0.134	idw	0.134	18	2.856	0.000	3.683	9.245	True	1.0	
nnk_	synth	0.2	mean	0.134	18	0.0	0.134	0.134	idw	0.134	18	2.856	0.000	3.683	8.840	True	1.0	
knn_	bci_iv	0.1	neare	2.535	6	9.404	2.478	2.608	trss	2.669	6	5.514	1.336	0.025	1.0	False	0.777	
knn_	bci_iv	0.2	neare	5.115	11	1.395	5.034	5.196	trss	5.261	11	9.730	1.453	0.025	1.0	False	0.570	
knn_	bci_iv	0.3	neare	9.129	11	1.440	9.037	9.196	trss	9.466	11	9.994	3.372	7.780	0.013	True	1.0	
gauss	physi	0.3	neare	8.304	26	6.861	8.052	8.555	idw	8.675	26	5.685	3.706	0.021	1.0	False	0.372	
knn_	physi	0.3	neare	8.520	6	0.0	8.520	8.520	idw	8.733	6	0.0	2.131	0.001	0.191	False	1.0	

Mejores combinaciones (ordenadas) — primeras filas

graph	dataset	single	best_test	mean	best_train	best_std	lower	upper	second	mean	second	delta	mean	p_raw	p_adj	significant	k_biserial	rank
knng	physio	0.3	nearest	8.520	6	0.0	8.520	8.520	idw	8.733	6	0.0	2.131	0.001	0.189	False	1.0	95
nnk	physio	0.3	nearest	8.520	7	0.0	8.520	8.520	gsmo	8.661	7	1.829	1.410	0.000	0.065	False	1.0	83
vknn	physio	0.3	nearest	8.520	6	0.0	8.520	8.520	idw	8.733	6	0.0	2.131	0.001	0.186	False	1.0	2
gauss	physio	0.4	rbfi_tf	5.691	4	6.013	5.641	5.737	rbfi_n	6.000	4	5.639	3.086	0.028	1.0	False	1.0	
gauss	physio	0.1	rbfi_tf	1.877	24	1.428	1.822	1.934	idw	1.921	25	1.479	4.449	0.340	1.0	False	0.16	
kalofc	physio	0.1	rbfi_tf	1.872	24	1.559	1.815	1.934	idw	1.972	25	1.425	1.004	0.013	1.0	False	0.413	33
aew	synth	0.1	spher	0.051	20	0.004	0.049	0.053	gsmo	0.051	60	0.007	0.000	0.480	1.0	False	0.106	66
aew	synth	0.2	spher	0.103	20	0.004	0.101	0.105	gsmo	0.104	60	0.008	0.001	0.110	1.0	False	0.24	
aew	synth	0.3	spher	0.182	20	0.003	0.180	0.183	gsmo	0.184	60	0.005	0.002	0.008	1.0	False	0.393	33
aew	synth	0.4	spher	0.233	20	0.006	0.230	0.235	gsmo	0.233	60	0.010	0.000	0.898	1.0	False	0.02	
gauss	synth	0.1	spher	0.051	15	0.004	0.049	0.053	tv	0.051	45	0.004	0.000	0.252	1.0	False	0.199	99
gauss	synth	0.2	spher	0.103	15	0.004	0.101	0.105	tv	0.103	45	0.004	0.000	0.745	1.0	False	0.057	77
gauss	synth	0.3	spher	0.182	15	0.003	0.180	0.184	tv	0.182	45	0.003	0.000	0.374	1.0	False	0.155	55
gauss	synth	0.4	spher	0.233	15	0.006	0.230	0.236	tv	0.233	45	0.006	0.000	0.669	1.0	False	0.075	55
kalofc	synth	0.1	spher	0.051	20	0.004	0.049	0.053	tv	0.051	60	0.004	7.864	0.426	1.0	False	0.12	
kalofc	synth	0.2	spher	0.103	20	0.004	0.101	0.105	tv	0.103	60	0.004	7.397	0.793	1.0	False	0.04	
kalofc	synth	0.3	spher	0.182	20	0.003	0.180	0.183	tv	0.182	60	0.003	7.536	0.759	1.0	False	0.046	66
kalofc	synth	0.4	spher	0.233	20	0.006	0.230	0.235	tv	0.233	60	0.006	8.569	0.426	1.0	False	0.12	
knng	synth	0.1	spher	0.051	20	0.004	0.049	0.053	tv	0.053	60	0.005	0.002	0.069	1.0	False	0.273	33
knng	synth	0.2	spher	0.103	20	0.004	0.101	0.105	tv	0.105	60	0.006	0.001	0.143	1.0	False	0.219	99
knng	synth	0.3	spher	0.182	20	0.003	0.180	0.183	tv	0.186	60	0.004	0.004	0.000	0.019	True	0.580	20
knng	synth	0.4	spher	0.233	20	0.006	0.230	0.235	tv	0.236	60	0.007	0.003	0.008	1.0	False	0.393	33
nnk	synth	0.1	spher	0.051	10	0.004	0.048	0.053	gsmo	0.055	30	0.004	0.004	0.029	1.0	False	0.466	66
nnk	synth	0.2	spher	0.103	10	0.004	0.100	0.106	gsmo	0.108	30	0.005	0.005	0.015	1.0	False	0.52	
nnk	synth	0.3	spher	0.182	10	0.003	0.180	0.184	gsmo	0.187	30	0.005	0.005	0.007	1.0	False	0.573	33
nnk	synth	0.4	spher	0.233	10	0.006	0.229	0.237	gsmo	0.239	30	0.008	0.006	0.010	1.0	False	0.546	66
vknn	synth	0.1	spher	0.051	20	0.004	0.049	0.053	tv	0.053	60	0.005	0.002	0.131	1.0	False	0.226	66
vknn	synth	0.2	spher	0.103	20	0.004	0.101	0.105	tv	0.105	60	0.006	0.001	0.184	1.0	False	0.199	99
vknn	synth	0.3	spher	0.182	20	0.003	0.180	0.183	tv	0.186	60	0.004	0.003	0.000	0.084	False	0.52	94
vknn	synth	0.4	spher	0.233	20	0.006	0.230	0.235	tv	0.235	60	0.007	0.002	0.026	1.0	False	0.333	33

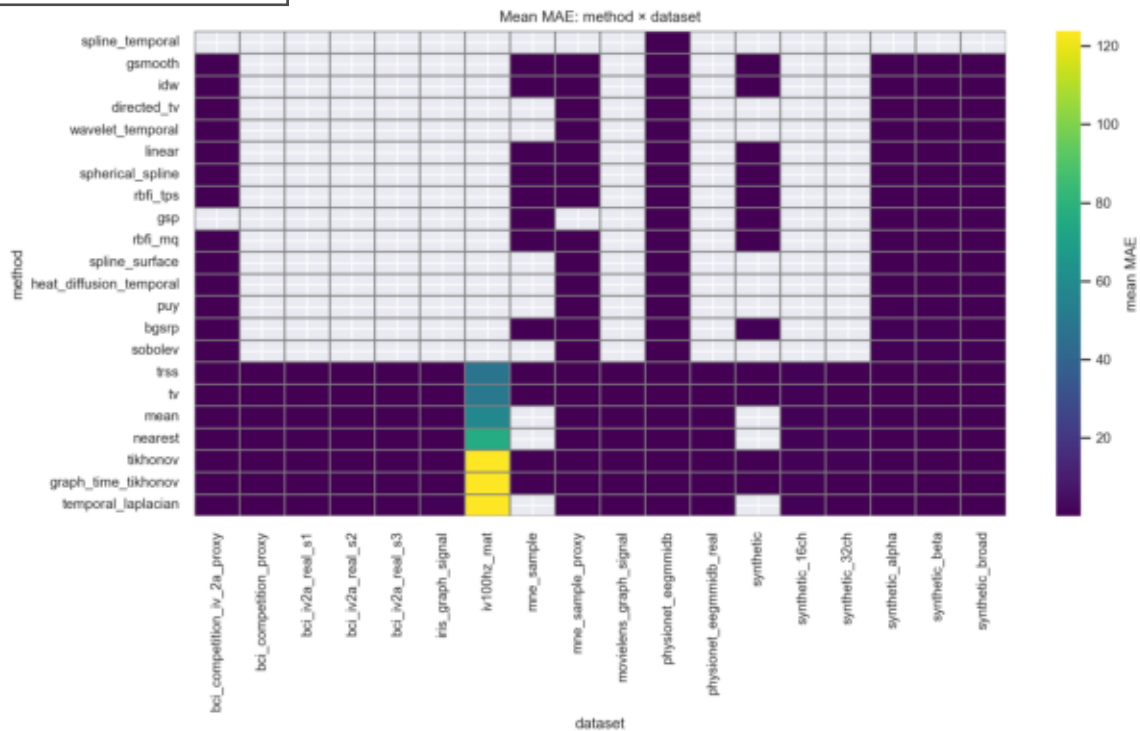
Mejores combinaciones (ordenadas) — primeras filas

graph	data	miss	single	best_test	mean	best_test	best_std	lower	upper	second	second_mean	second_std	delta_mean	p_raw	p_adj	significant	k_biseria
gauss	mne_	0.1	temp	1.298	1	nan	nan	nan	heat	1.313	1	nan	1.487	nan	nan	False	nan
gauss	mne_	0.3	temp	1.707	1	nan	nan	nan	tv	1.814	1	nan	1.079	nan	nan	False	nan
gauss	bci_cc	0.2	trss	4.180	10	5.262	3.880	4.501	direct	4.200	10	4.587	1.967	0.850	1.0	False	0.06
gauss	bci_cc	0.4	trss	5.404	10	8.378	4.936	5.948	direct	5.431	10	8.716	2.709	0.909	1.0	False	0.04
kalofc	bci_cc	0.2	trss	3.414	1	nan	nan	nan	tv	3.525	1	nan	1.110	nan	nan	False	nan
knn_	bci_cc	0.2	trss	4.248	13	6.311	3.948	4.598	temp	4.263	13	5.564	1.492	0.837	1.0	False	0.0532
knn_	bci_cc	0.3	trss	4.764	14	6.538	4.460	5.102	temp	4.852	14	2.740	8.737	0.395	1.0	False	0.1938
gauss	bci_cc	0.1	trss	6.995	19	2.829	6.873	7.117	tv	7.061	19	2.441	6.613	0.748	1.0	False	0.0637
gauss	bci_cc	0.2	trss	7.464	30	3.796	7.332	7.608	tv	7.544	30	3.702	7.999	0.371	1.0	False	0.1355
gauss	bci_cc	0.3	trss	7.778	33	3.976	7.646	7.918	tv	7.831	33	3.990	5.297	0.496	1.0	False	0.0982
gauss	bci_cc	0.4	trss	7.835	13	1.807	7.743	7.924	tv	8.099	13	1.238	2.639	0.000	0.133	False	0.7751
kalofc	bci_cc	0.1	trss	7.148	9	3.269	6.943	7.341	tv	7.525	9	4.425	3.777	0.093	1.0	False	0.4814
kalofc	bci_cc	0.2	trss	7.524	16	3.009	7.389	7.671	tv	7.669	16	2.977	1.448	0.136	1.0	False	0.3125
kalofc	bci_cc	0.3	trss	7.928	16	3.110	7.782	8.073	tv	8.047	16	2.538	1.186	0.180	1.0	False	0.2812
kalofc	bci_cc	0.4	trss	7.899	3	2.073	7.744	8.135	tv	8.023	3	2.331	1.241	0.7	1.0	False	0.3333
knn_	bci_cc	0.1	trss	6.706	26	2.276	6.622	6.786	tv	6.798	26	2.424	9.158	0.237	1.0	False	0.1923
knn_	bci_cc	0.2	trss	7.095	37	3.327	6.992	7.198	tv	7.180	37	3.129	8.525	0.330	1.0	False	0.1322
knn_	bci_cc	0.3	trss	7.436	40	3.591	7.328	7.546	tv	7.515	40	3.514	7.902	0.279	1.0	False	0.1412
knn_	bci_cc	0.4	trss	7.485	16	2.136	7.390	7.588	tv	7.679	16	2.186	1.935	0.016	1.0	False	0.5
gauss	bci_iv	0.1	trss	8.470	10	3.326	8.289	8.669	tv	1.253	10	4.313	4.068	0.000	0.029	True	1.0
gauss	bci_iv	0.2	trss	1.672	12	6.436	1.636	1.704	tv	2.406	12	9.097	7.334	3.462	0.005	True	1.0
gauss	bci_iv	0.3	trss	3.274	10	9.145	3.220	3.325	tv	4.407	10	1.091	1.133	0.000	0.029	True	1.0
gauss	bci_iv	0.4	trss	4.353	2	5.686	4.313	4.393	tv	5.674	2	6.239	1.320	nan	nan	False	nan
knn_	bci_iv	0.1	trss	8.554	36	4.322	8.410	8.684	tv	1.252	36	5.109	3.970	2.602	7.442	True	1.0
knn_	bci_iv	0.2	trss	1.589	64	2.632	1.517	1.648	mean	2.406	58	8.928	8.165	1.303	3.793	True	1.0
knn_	bci_iv	0.3	trss	3.217	41	7.728	3.193	3.240	tv	4.393	41	1.029	1.176	5.542	1.596	True	1.0
knn_	bci_iv	0.4	trss	4.253	2	6.381	4.248	4.257	tv	5.673	2	6.232	1.420	nan	nan	False	nan
gauss	bci_iv	0.1	trss	2.466	4	8.633	2.394	2.539	neare	3.282	4	1.722	8.155	0.028	1.0	False	1.0
gauss	bci_iv	0.2	trss	5.118	6	1.164	5.029	5.189	neare	6.649	4	2.088	1.531	0.013	1.0	False	1.0
gauss	bci_iv	0.3	trss	9.477	4	1.496	9.331	9.580	neare	1.193	4	2.734	2.455	0.028	1.0	False	1.0

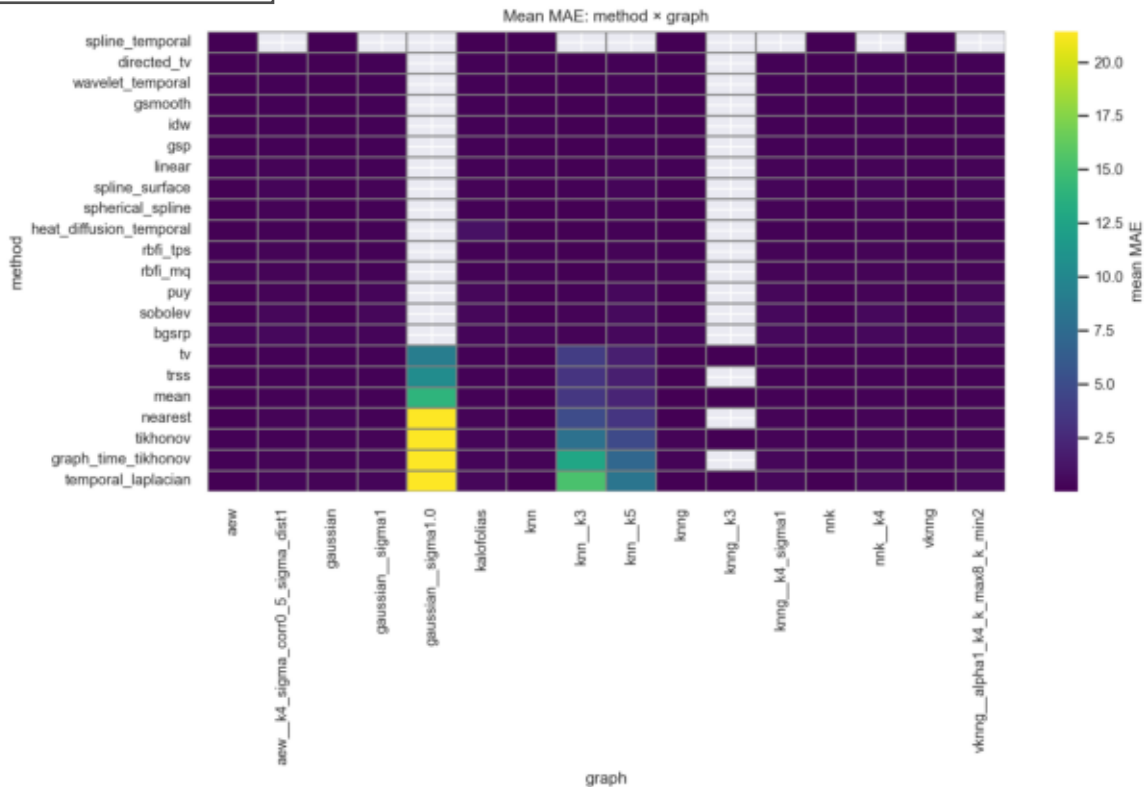
Mejores combinaciones (ordenadas) — primeras filas

graph	dataset	single	best_test	mean	best_n	best_std	lower	upper	second	mean	second	delta	mean	p_raw	p_adj	significant	k_biseria
gauss	bci_iv	0.4	trss	1.262	2	2.078	1.247	1.277	tv	1.590	2	2.031	3.277	nan	nan	False	nan
knn	bci_iv	0.1	trss	2.911	10	1.098	2.849	2.977	nearest	3.338	10	1.911	4.267	0.000	0.028	True	1.0
knn	bci_iv	0.2	trss	5.929	22	1.608	5.861	5.998	nearest	6.629	18	1.919	6.998	6.672	1.321	True	1.0
knn	bci_iv	0.3	trss	1.062	15	2.073	1.052	1.072	nearest	1.193	15	2.928	1.306	3.113	0.000	True	1.0
knn	bci_iv	0.4	trss	1.394	2	4.205	1.364	1.424	tv	1.590	2	2.033	1.956	nan	nan	False	nan
gauss	iris_gr	0.1	trss	0.102	10	0.008	0.097	0.107	tv	0.122	10	0.012	0.020	0.002	0.387	False	0.8
gauss	iris_gr	0.2	trss	0.102	17	0.008	0.098	0.106	tv	0.123	17	0.011	0.020	2.454	0.004	True	0.847
gauss	iris_gr	0.3	trss	0.102	15	0.008	0.097	0.106	tv	0.122	15	0.011	0.020	0.000	0.025	True	0.813
gauss	iris_gr	0.4	trss	0.228	2	0.007	0.223	0.233	tv	0.266	2	0.008	0.038	nan	nan	False	nan
knn	iris_gr	0.1	trss	0.105	28	0.008	0.102	0.108	tv	0.126	28	0.011	0.020	1.200	2.220	True	0.823
knn	iris_gr	0.2	trss	0.107	114	0.029	0.102	0.113	tv	0.134	114	0.023	0.027	6.337	1.837	True	0.719
knn	iris_gr	0.3	trss	0.105	21	0.009	0.101	0.109	tv	0.126	21	0.011	0.020	8.844	0.001	True	0.800
knn	iris_gr	0.4	trss	0.253	2	0.006	0.249	0.257	mean	0.268	2	0.011	0.014	nan	nan	False	nan
knn	iris_gr	0.2	trss	0.103	5	0.009	0.096	0.111	tv	0.127	5	0.011	0.023	0.015	1.0	False	0.92
knn	iris_gr	0.3	trss	0.103	5	0.009	0.096	0.111	tv	0.127	5	0.011	0.023	0.015	1.0	False	0.92
knn	iv100	0.1	trss	16.15	20	0.385	15.99	16.32	tv	17.51	20	1.043	1.360	8.300	0.001	True	0.825
knn	iv100	0.2	trss	36.77	118	15.14	33.97	39.55	tv	46.32	118	12.07	9.549	2.948	8.520	True	0.636
knn	iv100	0.3	trss	65.71	25	0.917	65.36	66.07	mean	73.18	25	0.708	7.470	1.304	3.601	True	1.0
knn	iv100	0.4	trss	88.19	2	0.291	87.98	88.39	mean	94.15	2	1.017	5.966	nan	nan	False	nan
knn	iv100	0.3	trss	62.75	5	1.007	61.98	63.52	tv	63.54	5	5.371	0.792	0.841	1.0	False	0.12

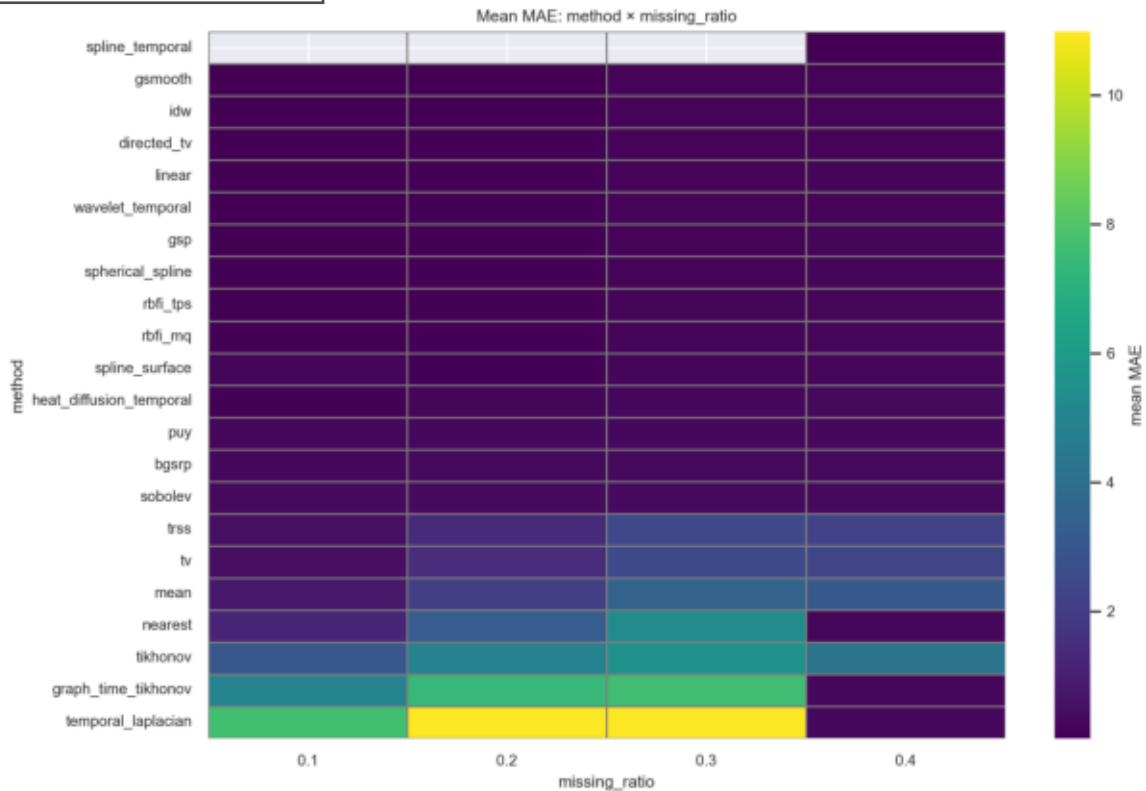
heatmap_method_dataset_mean_mae.png



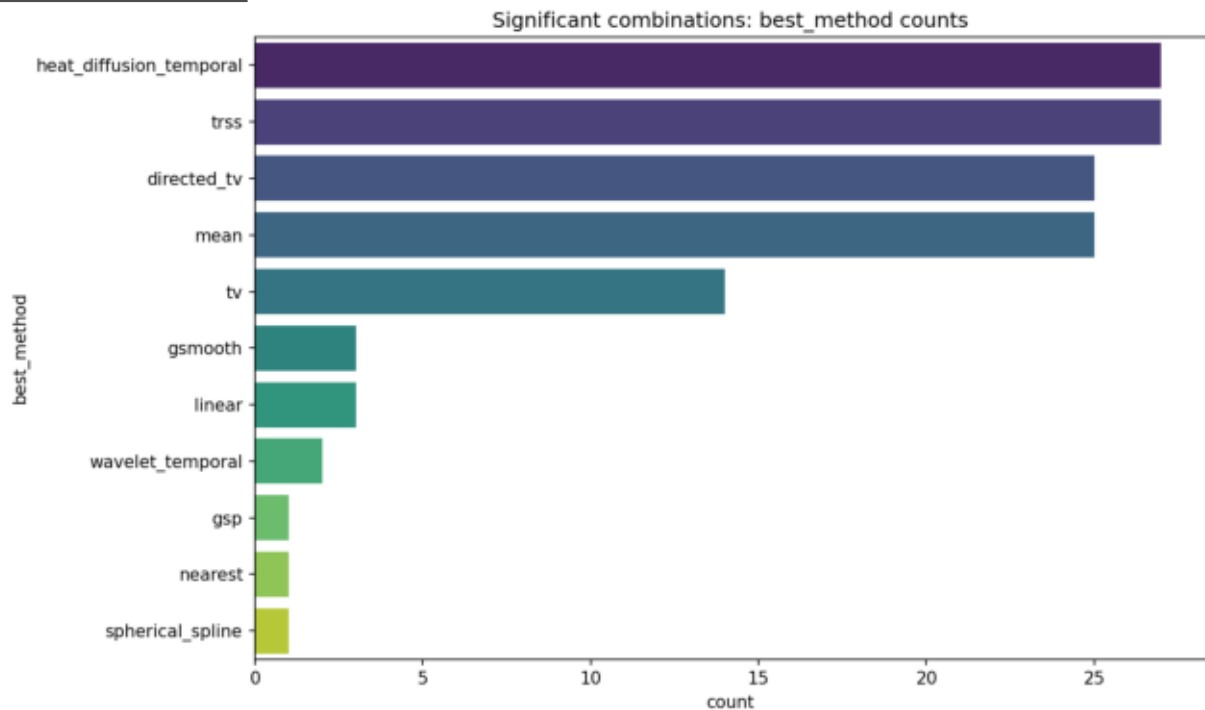
heatmap_method_graph_mean_mae.png

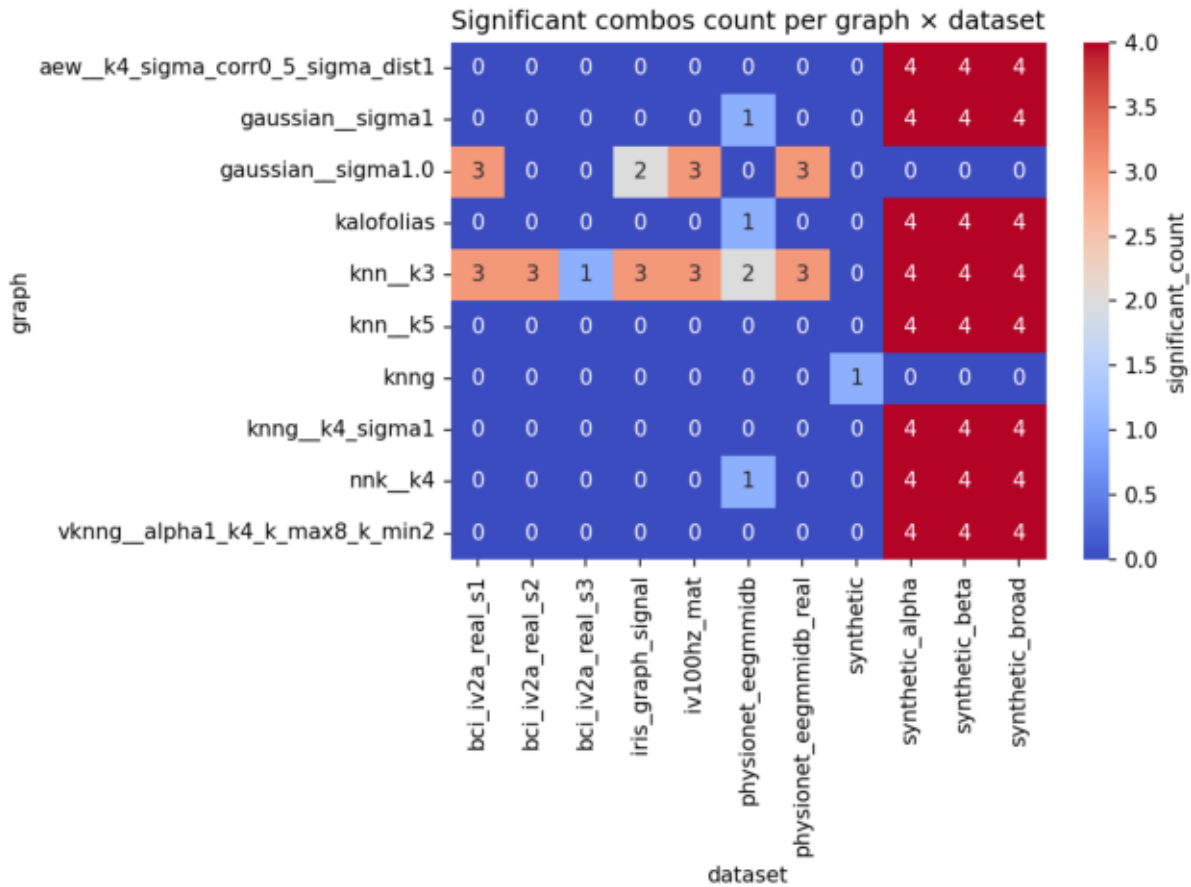


heatmap_method_missing_ratio_mean_mae.png

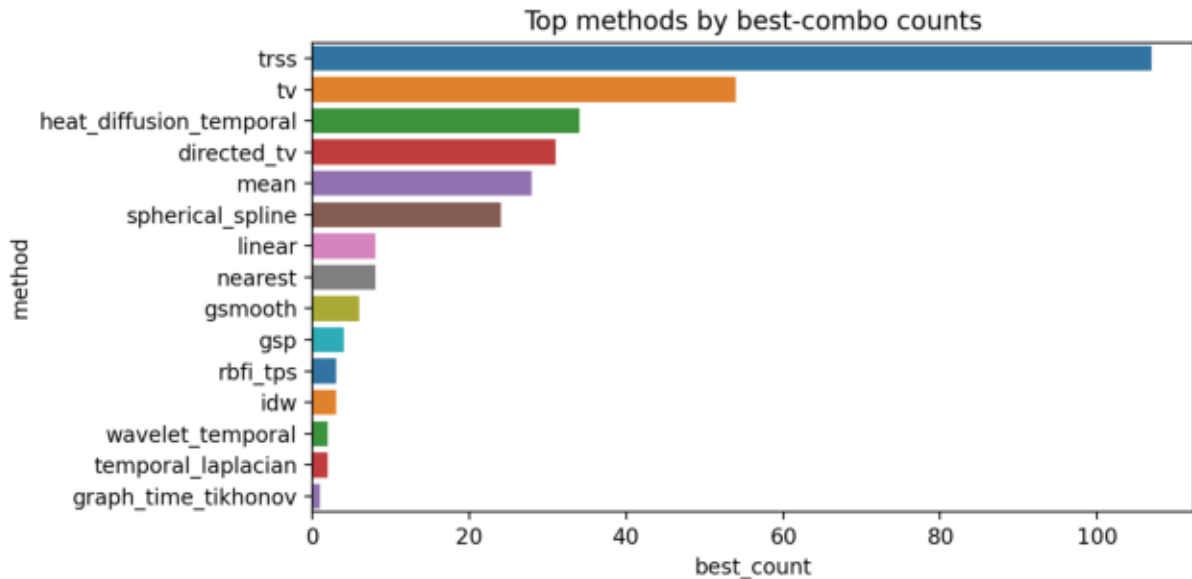


significant_best_method_counts.png





top_best_methods_overall.png



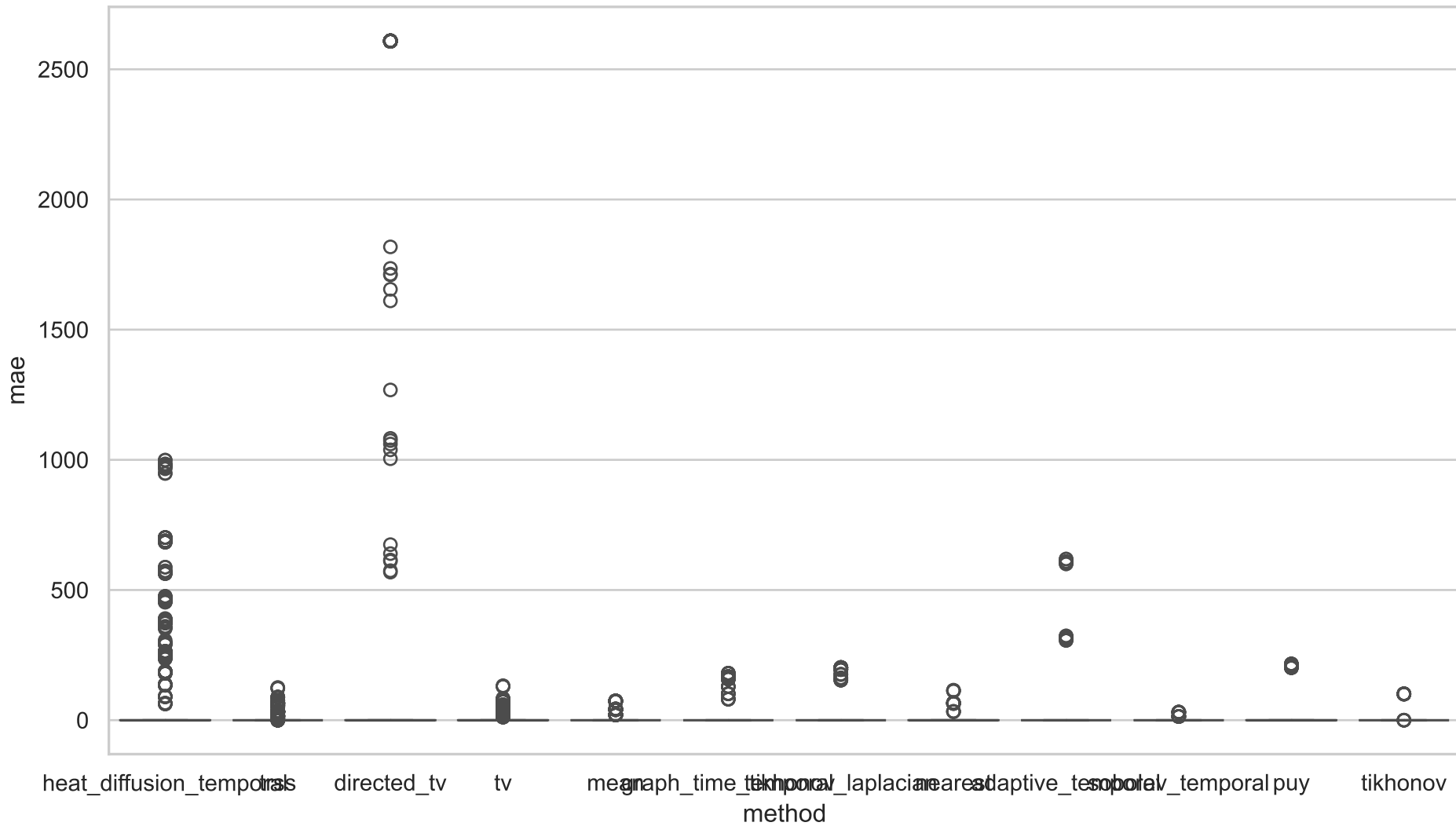
Archivos incluidos (índice)

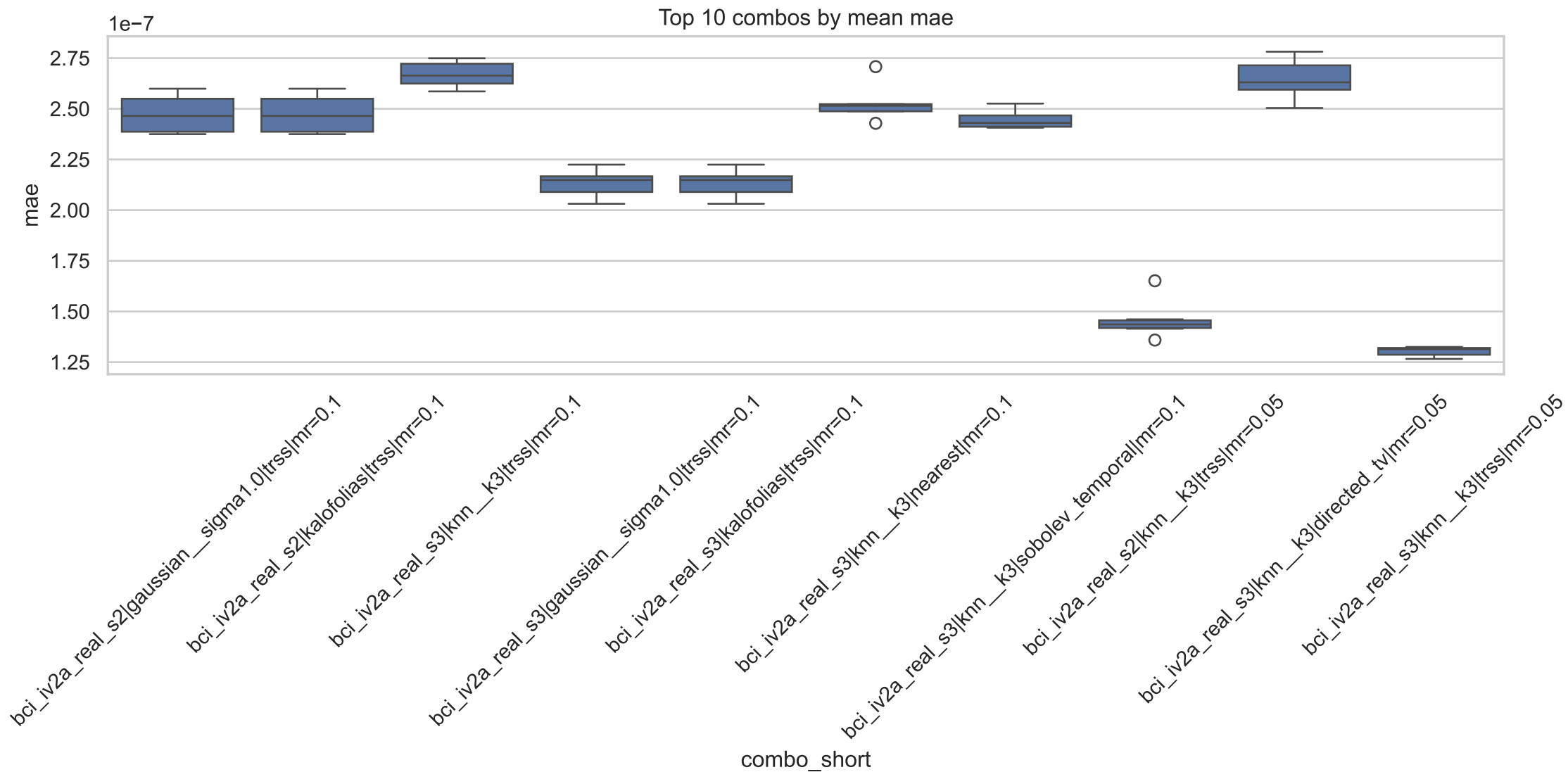
results\analysis\batches\batch_001_agg.csv
results\analysis\batches\batch_001_box_frontal_band.png
results\analysis\batches\batch_001_box_left_lateral_temporal.png
results\analysis\batches\batch_001_box_midline_central.png
results\analysis\batches\batch_001_box_occipital_band.png
results\analysis\batches\batch_001_box_right_lateral_temporal.png
results\analysis\batches\batch_001_consolidated.csv
results\analysis\batches\batch_001_stats.json
results\analysis\batches\batch_002_agg.csv
results\analysis\batches\batch_002_box_None.png
results\analysis\batches\batch_002_consolidated.csv
results\analysis\batches\batch_002_stats.json
results\analysis\batches\batch_003_agg.csv
results\analysis\batches\batch_003_box_None.png
results\analysis\batches\batch_003_consolidated.csv
results\analysis\batches\batch_003_stats.json
results\analysis\batches\batch_004_agg.csv
results\analysis\batches\batch_004_box_None.png
results\analysis\batches\batch_004_consolidated.csv
results\analysis\batches\batch_004_stats.json
results\analysis\batches\batch_005_agg.csv
results\analysis\batches\batch_005_box_None.png
results\analysis\batches\batch_005_consolidated.csv
results\analysis\batches\batch_005_stats.json
results\analysis\batches\batch_006_agg.csv
results\analysis\batches\batch_006_box_None.png
results\analysis\batches\batch_006_consolidated.csv
results\analysis\batches\batch_006_stats.json
results\analysis\batches\batch_007_agg.csv
results\analysis\batches\batch_007_box_None.png
results\analysis\batches\batch_007_consolidated.csv
results\analysis\batches\batch_007_stats.json
results\analysis\batches\batch_008_agg.csv
results\analysis\batches\batch_008_box_None.png
results\analysis\batches\batch_008_consolidated.csv
results\analysis\batches\batch_008_stats.json
results\analysis\batches\batch_009_agg.csv
results\analysis\batches\batch_009_box_None.png
results\analysis\batches\batch_009_consolidated.csv
results\analysis\batches\batch_009_stats.json
results\analysis\batches\batch_010_agg.csv
results\analysis\batches\batch_010_box_None.png
results\analysis\batches\batch_010_consolidated.csv
results\analysis\batches\batch_010_stats.json
results\analysis\batches\batch_011_agg.csv
results\analysis\batches\batch_011_box_None.png
results\analysis\batches\batch_011_consolidated.csv
results\analysis\batches\batch_011_stats.json
results\analysis\batches\batch_012_agg.csv
results\analysis\batches\batch_012_box_None.png
results\analysis\batches\batch_012_consolidated.csv
results\analysis\batches\batch_012_stats.json
results\analysis\batches\batch_013_agg.csv
results\analysis\batches\batch_013_box_None.png
results\analysis\batches\batch_013_consolidated.csv
results\analysis\batches\batch_013_stats.json
results\analysis\batches\batch_014_agg.csv
results\analysis\batches\batch_014_box_None.png
results\analysis\batches\batch_014_consolidated.csv
results\analysis\batches\batch_014_stats.json
results\analysis\batches\batch_015_agg.csv
results\analysis\batches\batch_015_box_None.png
results\analysis\batches\batch_015_consolidated.csv
results\analysis\batches\batch_015_stats.json
results\analysis\batches\best_by_combo_sorted.csv
results\analysis\batches\best_method_by_combo.csv
results\analysis\batches\best_method_by_combo_stats.csv
results\analysis\batches\best_method_counts.csv
results\analysis\batches\consolidated_all_batches.csv
results\analysis\batches\figures\heatmap_best_method_by_graph_counts.png
results\analysis\batches\figures\heatmap_method_dataset_mean_mae.png
results\analysis\batches\figures\heatmap_method_graph_mean_mae.png
results\analysis\batches\figures\heatmap_method_missing_ratio_mean_mae.png
results\analysis\batches\figures\significant_best_method_counts.png
results\analysis\batches\figures\significant_heatmap_graph_dataset.png
results\analysis\batches\figures\top_best_methods_overall.png
results\analysis\batches\global_by_combo.csv
results\analysis\batches\global_by_dataset.csv
results\analysis\batches\global_by_graph.csv
results\analysis\batches\global_by_iteration.csv

Archivos incluidos (índice)

results\analysis\batches\global_by_method.csv
results\analysis\batches\global_by_method_graph.csv
results\analysis\batches\global_by_missing_ratio.csv
results\analysis\batches\global_by_scenario.csv
results\analysis\batches\global_method_stats.csv
results\analysis\batches\global_summary.md
results\analysis\batches\interactive\best_method_counts.html
results\analysis\batches\interactive\heatmap_dataset.html
results\analysis\batches\interactive\heatmap_graph.html
results\analysis\batches\interactive\heatmap_missing.html
results\analysis\batches\interactive\index.html
results\analysis\batches\overall_method_ranking.csv
results\analysis\batches\pivot_best_method_by_graph_counts.csv
results\analysis\batches\pivot_method_by_dataset_mean_mae.csv
results\analysis\batches\pivot_method_by_graph.csv
results\analysis\batches\pivot_method_by_graph_mean_mae.csv
results\analysis\batches\pivot_method_by_missing_ratio_mean_mae.csv
results\analysis\batches\readable_best_summary.txt
results\analysis\batches\report_summary.pdf
results\analysis\batches\significant_combinations.csv
results\analysis\batches\significant_combinations.xlsx
results\analysis\batches\summary_report.md
results\analysis\batches\summary_report.xlsx
results\analysis\batches\top20_delta_combos.csv
results\analysis\batches\top20_methods_by_mean_mae.csv
results\analysis\batches\top3_by_block_counts.csv
results\analysis\batches\top3_by_dataset_counts.csv
results\analysis\batches\top_method_counts.csv

Distribution of mae by method (it131-it150)





Worst 10 combos by mean mae

